

Mini-ATS (Application Tracking System)

Executive Summary

The Mini-ATS is a digital tool designed to replace messy spreadsheets and emails during the hiring process. It allows candidates to apply for jobs and gives the hiring team a "bird's-eye view" of where every candidate stands in the hiring process.

Key Features

A. Job Management (The "Postings")

- **Create & Edit Jobs:** Recruiters can post new roles
- **Capacity Tracking:** Keep track of how many openings are available for a specific role.
- **Status Control:** Open or close a job posting with one click.

B. Candidate Management (The "Profiles")

- **Digital Database:** Stores name, email, phone, and a link to the resume.
- **Resume Intelligence:** The system automatically "reads" the resume to pull out skills and education so you don't have to hunt for them.

C. Hiring Pipeline (The "Stages")

- **Custom Progress:** Instead of a simple "Yes/No," candidates move through:
 1. *HR Screening / Assignment* (Optional)
 2. *Technical Round* (Mandatory)
 3. *HR Round* (Mandatory)
 4. *Founder Round* (Mandatory)
- **Flexibility:** The system allows you to jump stages (e.g., moving a star candidate straight to the Technical Round).

D. Automation & Notifications (The "Follow-up")

- **Instant Alerts:** When a recruiter moves a candidate to the next round, the system can automatically email the candidate to let them know.

Flow Diagram

1. **Job Created:** Recruiter posts a role.
2. **Candidate Applies:** Form is filled out; resume is uploaded to Google Drive.
3. **Review:** Recruiter sees all applicants in a "Dashboard."
4. **Movement:** Recruiter drags a candidate from "Applied" to "Technical Round."
5. **Alert:** Candidate gets an email; the system logs the time and date of the move.

System Assumptions & Decisions

- **User Types:** There are two types of users: **Candidates** (who only see the application form) and **Team Members/HR** (who see the tracking dashboard).
- **Storage:** each and every things we will store in from of table in postgresql and resume link will be stored in form of String.

- **Simplicity:** The system focuses on the interview rounds rather than scheduling (calendars).

API & Data Structure (Simplified)

Feature	Key Information Collected
Job Details	Title, Description, Total Openings.
Candidate Info	Name, Contact Info, Extracted Skills.
The Tracking Log	Who moved the candidate? From which stage to which stage? When?

API Structure (How the Front & Back Talk):

❖ Auth & User Management

- POST /auth/login – HR enters credentials to receive a **JWT Token**.
- GET /auth/me – Returns the current logged-in HR user's details.

❖ Job Management (HR Only)

- GET /jobs – (Public) So candidates can see what to apply for.
- POST /jobs – **(HR Only)** Create a role.
- PUT /jobs/{id} – **(HR Only)** Update description or close a role.

❖ Candidate & Pipeline (Hybrid)

- POST /apply/{job_id} – (Public) Candidate submission.
- GET /applications – **(HR Only)** The main dashboard data.
- PATCH /applications/{id}/status – **(HR Only)** Change the round (e.g., Technical $\rightarrow$ HR Round).

❖ Frontend Overview:

1. Candidate View (The Public Facing Side)

This should be clean and mobile-responsive.

- **Job Listing Page:**
 - Cards displaying Job Title, Location (Remote/Office), and a "View Details" button.
- **Application Form:**
 - Fields for Name, Email, Phone.
 - A simple text input for the **Resume Link**.
 - A "Submit" button that provides instant feedback (e.g., "Application Received!")

2. HR/Recruiter Dashboard (The Internal Side)

This is where the complex data lives. You need three main components:

A. The Pipeline Board

This is the heart of the app. Imagine columns representing your stages:

- **Columns:** Screening | Technical | HR Round | Founder Round
- **Cards:** Each card represents a candidate.
- **Action:** Drag a card from one column to another to trigger the stage update API.

B. Candidate Detail Modal

When HR clicks a candidate's card, a side-panel or modal opens showing:

- **Contact Info:** Quick links to email or call.
- **Resume Link:** A clickable button to open the Google Drive link in a new tab.
- **Extracted Data:** A summary section showing "Top Skills" and "Education" (the data parsed by your backend).
- **Timeline/History:** A list showing when they moved through stages (e.g., *"Moved to Technical Round by Admin on April 28"*).

C. Job Management Page

- A table showing all jobs.
- Stats: "Active Applications," "Openings Left."
- "Create New Job" button that opens a form

Schemas:

Table Name	Purpose	Key Columns
users	Stores HR/Admin credentials for login.	id, username, password_hash, role (e.g., 'HR')
jobs	Stores the details of the open positions.	id, title, description, openings_count, is_active
candidates	Stores the personal info of people applying.	id, full_name, email, phone, resume_link (String)
applications	The "link" between a candidate and a job.	id, candidate_id, job_id, current_stage, status
application_history	Tracks every move (for the tracking UI).	id, application_id, from_stage, to_stage, moved_at

Visualizing the Relationships

- **One-to-Many:** One **Job** can have many **Applications**.
- **One-to-Many:** One **Candidate** can have many **Applications**.
- **One-to-Many:** One **Application** has many **History entries** (tracking the movement).

Tech Stack:

Layer	Technology	Why?
Language	Python 3.10+	Best for data extraction and fast API dev.
API Framework	FastAPI	High performance and easy to learn.
Frontend	React + Vite	Vite makes the build process incredibly fast.
Database	PostgreSQL	Reliability and complex relationship handling.
Styling	Tailwind CSS	Rapid UI development for the HR Dashboard.

Scope Decisions

- **In-Scope:** Managing the "Flow" of people through the four rounds.
- **Out-of-Scope:** We are not building a video calling tool or a background check system yet.